

DRIVEVALUE AI

From **Data** to **Decisions** —
An Explainable Machine Learning
Framework for Intelligent
Used Vehicle Price Prediction



DriveValue AI

From Data to Decisions

An Explainable Machine Learning Framework for Intelligent Used Vehicle Price Prediction

Capstone Project Report

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At-a-glance

Source: Indian online vehicle marketplace (all prices in ₹)

Target variable: **AskPrice (₹)**

Original dataset: **24,575** records, **11** features

Final analytical dataset: **14,988** cleaned records, **466** engineered features

Executive Summary

Every year, millions of used vehicles are bought and sold worldwide, yet determining a fair asking price remains one of the most subjective decisions within the automotive industry. Similar vehicles are

frequently advertised at vastly different prices due to inconsistent valuation practices, human judgement, and rapidly changing market conditions.

For vehicle dealerships, online marketplaces, and financial institutions, inaccurate pricing carries significant business consequences. Overpriced vehicles remain unsold for longer periods, reducing inventory turnover, while underpriced vehicles lead to unnecessary revenue loss. Traditional pricing methods rely heavily on experience rather than evidence, creating inconsistencies that reduce customer confidence and operational efficiency.

DriveValue AI was developed as an explainable machine learning framework capable of predicting used vehicle asking prices from historical vehicle listings. Rather than replacing human expertise, the system augments pricing decisions with objective, data-driven insights while maintaining transparency through Explainable Artificial Intelligence (XAI).

The project followed the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology, progressing through business understanding, exploratory data analysis, feature engineering, predictive modelling, evaluation, explainability, and deployment. After extensive preprocessing, the final dataset contained 14,988 cleaned vehicle listings represented by 466 engineered features.

Three regression algorithms were evaluated: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. Comparative evaluation identified the Random Forest Regressor as the strongest-performing model.

Final production model performance (Random Forest Regressor)

Mean Absolute Error (MAE): **₹398,156**

Root Mean Squared Error (RMSE): **₹1,267,201**

R² Score: **0.446**

To improve stakeholder trust, SHAP (SHapley Additive Explanations) was integrated to explain how vehicle characteristics influenced individual predictions, transforming the model from a “black box” into an interpretable decision-support tool. Finally, the model was deployed as an interactive Streamlit web application, demonstrating how users can generate real-time vehicle price estimates through a simple graphical interface.

DriveValue AI demonstrates how machine learning can support more transparent, consistent, and evidence-based pricing decisions within the automotive industry while providing a scalable foundation for future intelligent valuation systems.

1. Introduction

“Pricing a used vehicle should not be a guessing game.”

Every day, thousands of used vehicles are listed on online marketplaces and dealership websites. Although these vehicles often share similar characteristics, their asking prices frequently differ considerably. Such inconsistencies arise because pricing decisions are commonly based on personal

judgement, previous experience, or informal market comparisons rather than objective analytical methods.

The rapid growth of digital vehicle marketplaces has significantly increased both the volume and complexity of pricing decisions. Sellers must consider numerous variables including vehicle age, mileage, fuel type, ownership history, transmission type, brand reputation, and model popularity when determining an appropriate asking price. Evaluating all these variables simultaneously becomes increasingly difficult as inventory size grows.

Machine learning provides an opportunity to improve this process by learning complex relationships directly from historical vehicle data. Rather than relying solely on intuition, predictive algorithms can identify hidden patterns that produce more consistent pricing recommendations while reducing human bias. However, predictive accuracy alone is insufficient: business users must also understand *why* a model recommends a particular price before trusting its predictions. This creates a need for Explainable Artificial Intelligence (XAI), allowing model decisions to be interpreted and validated by human stakeholders.

DriveValue AI was therefore developed to combine predictive performance with explainability, producing an intelligent pricing framework that supports transparent and data-driven vehicle valuation.

Project Aim

The aim of this project was to develop an explainable machine learning framework capable of predicting used vehicle asking prices using historical vehicle listing data while providing transparent explanations for each prediction.

Objectives

The project sought to:

- develop an end-to-end predictive pricing framework;
 - compare multiple regression algorithms;
 - identify the most suitable predictive model;
 - explain model behaviour using SHAP;
 - demonstrate deployment through a Streamlit application; and
 - provide practical recommendations for real-world implementation.
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2. Business Understanding and Data Preparation

Transforming Raw Vehicle Listings into a Machine Learning Dataset

Developing an accurate vehicle price prediction model required more than simply selecting a machine learning algorithm. The quality of the predictions depended heavily on the quality of the data used

during training. Consequently, a significant portion of this project focused on understanding, cleaning, and preparing the dataset before model development.

The project utilised a real-world dataset containing historical used vehicle listings collected from an Indian online vehicle marketplace. Each record represented an individual vehicle advertisement and contained information describing both the vehicle and the listing itself. Variables included the vehicle brand, model, manufacturing year, vehicle age, kilometres driven, transmission type, fuel type, ownership history, advertisement posting date, and the advertised asking price.

Note — The source dataset originates from an Indian vehicle marketplace; therefore, all prices reported throughout this report are expressed in Indian Rupees (₹).

Dataset Overview

Following the data understanding phase, the original dataset was inspected to identify data quality issues that could negatively affect model performance. Initial exploration revealed duplicated observations, missing values, inconsistent categorical information, and formatting inconsistencies across several variables.

Dataset characteristic	Value
Original records	24,575
Final cleaned records	14,988
Original features	11
Engineered features	466
Target variable	AskPrice (₹)

Table 1: Summary of the dataset after preprocessing.

3. Data Cleaning

A structured preprocessing pipeline was implemented to improve data quality before model training. The cleaning process included:

- Removing duplicate vehicle listings.
- Handling missing values through appropriate imputation strategies.
- Standardising inconsistent categorical values.
- Converting numerical variables into appropriate data types.
- Removing unnecessary information that did not contribute to predictive performance.

- Preparing categorical variables for machine learning through one-hot encoding.

These preprocessing steps ensured that the final dataset contained complete, consistent, and machine-learning-ready observations.

4. Feature Engineering

Feature engineering played a critical role in improving the predictive capability of the final model.

Categorical variables such as vehicle brand, model, transmission type, fuel type, ownership history, and advertisement posting date were transformed into numerical representations using one-hot encoding. This process enabled the machine learning algorithms to interpret categorical information while preserving relationships between different vehicle characteristics.

Numerical variables—including Year, Age, and Kilometres Driven—were retained as continuous predictors because they provide important information regarding vehicle depreciation and usage.

Following feature engineering, the dataset expanded from 11 original variables to 466 predictive features, providing a richer representation of each vehicle listing for model training.

Data Preparation Outcome

The completed preprocessing pipeline produced a high-quality analytical dataset suitable for predictive modelling. By removing duplicate observations, appropriately handling missing values, standardising inconsistent information, and engineering informative features, the dataset became substantially more reliable for machine learning.

The resulting dataset of 14,988 cleaned vehicle listings and 466 engineered features provided a robust foundation for model development and comparative evaluation. This preparation stage was instrumental in enabling the subsequent machine learning models to identify meaningful relationships between vehicle characteristics and asking prices while improving overall predictive performance.

5. Model Development and Evaluation

Developing the Predictive Pricing Model

Following the successful preparation of the dataset, the next phase focused on developing machine learning models capable of accurately predicting used vehicle asking prices. Rather than relying on a single algorithm, multiple regression models were developed and evaluated to identify the approach that best captured the complex relationships between vehicle characteristics and market prices.

The cleaned dataset was divided into independent variables (predictor features) and the target variable (AskPrice). To evaluate model performance on previously unseen data, the dataset was

partitioned into training and testing subsets using an 80:20 train-test split. This ensured that model performance reflected its ability to generalise beyond the training data rather than simply memorising historical observations.

The evaluation process followed a comparative modelling strategy in which multiple regression algorithms were trained under identical conditions before selecting the best-performing model for deployment.

Candidate Machine Learning Models

Three supervised machine learning regression algorithms were evaluated during model development:

- **Linear Regression** — implemented as the baseline model due to its simplicity and interpretability, with the assumption of linear relationships between predictors and AskPrice. While computationally efficient, it is often unable to capture complex nonlinear relationships common in vehicle pricing data.
- **Decision Tree Regressor** — developed to model nonlinear relationships by recursively partitioning the feature space into decision regions. Without sufficient regularisation, decision trees are susceptible to overfitting, which can reduce generalisation to unseen listings.
- **Random Forest Regressor** — an ensemble approach combining predictions from multiple decision trees to improve accuracy and reduce variance. By aggregating independently trained trees, Random Forest produces more stable and reliable predictions and is well suited to datasets containing both numerical and categorical variables.

Model Selection

Each regression algorithm was evaluated using the same testing dataset to ensure a fair comparison. Performance was assessed using three widely accepted regression evaluation metrics:

- Mean Absolute Error (MAE), measuring the average prediction error.
- Root Mean Squared Error (RMSE), measuring the magnitude of prediction errors while penalising larger deviations.
- Coefficient of Determination (R^2), measuring the proportion of variation in vehicle prices explained by the model.

Among the evaluated algorithms, the Random Forest Regressor consistently produced the strongest predictive performance. The model demonstrated superior ability to capture complex nonlinear relationships between vehicle characteristics and asking prices while maintaining better generalisation than the alternative models. Consequently, the Random Forest Regressor was selected as the final production model and used throughout the remainder of the project.

Hyperparameter Optimisation

Following initial model comparison, the selected Random Forest model was further investigated using Randomized Search Cross-Validation to determine whether performance could be improved through hyperparameter optimisation. Multiple combinations of model parameters, including the number of trees, maximum tree depth, minimum samples required for splitting, minimum samples per leaf, and feature selection strategy were evaluated using cross-validation.

Although the optimisation process successfully identified an alternative parameter configuration, evaluation on the independent testing dataset demonstrated that the tuned model did not outperform the original Random Forest model. To prioritise model stability and generalisation, the original Random Forest model was retained as the final production model. This decision reflects good machine learning practice by selecting the model that performs best on unseen data rather than simply choosing the model with the most complex parameter configuration.

6. Final Model Performance

The selected Random Forest production model achieved the following performance on the testing dataset:

Metric	Result
Mean Absolute Error (MAE)	₹398,156
Root Mean Squared Error (RMSE)	₹1,267,201
R ² Score	0.446

Table 2: Final model performance (Random Forest Regressor).

These results indicate that the model successfully captured a substantial proportion of the variation in used vehicle asking prices while maintaining acceptable prediction error across previously unseen vehicle listings. Although larger prediction errors remained for certain luxury and premium vehicles, overall performance demonstrated that the Random Forest model provides a reliable foundation for intelligent pricing recommendations.

Model Evaluation

To further assess predictive performance, the relationship between actual vehicle prices and predicted prices was visualised using an Actual versus Predicted scatter plot. The majority of observations clustered around the diagonal reference line, demonstrating strong agreement between actual and predicted prices. The concentration of observations around this line indicates that the model performs particularly well within the mainstream vehicle market, where most listings occur.

Residual analysis further confirmed that prediction errors were randomly distributed around zero, suggesting that the model does not consistently overestimate or underestimate vehicle prices. The

absence of systematic residual patterns indicates good generalisation and supports the suitability of the Random Forest model for deployment.

7. Explainable AI and Deployment

Model Explainability and Deployment Readiness

While predictive accuracy is an essential characteristic of a successful machine learning model, modern artificial intelligence systems are increasingly expected to provide transparent and interpretable predictions. In practical business environments, stakeholders require an understanding of why a particular prediction was generated before they are willing to trust or adopt automated decision-making systems.

To address this challenge, DriveValue AI incorporates Explainable Artificial Intelligence (XAI) through the use of SHapley Additive Explanations (SHAP). In addition to producing accurate vehicle price predictions, the project also demonstrates how individual predictions can be explained and interpreted, thereby increasing stakeholder confidence and supporting responsible AI practices. Beyond explainability, the project was extended to include deployment planning and a fully functional web-based prototype, demonstrating how the trained machine learning model could be integrated into a real-world vehicle pricing environment.

Explainable Artificial Intelligence (SHAP)

Traditional machine learning algorithms such as Random Forest are often considered “black-box” models because their internal decision-making processes are not immediately interpretable. Although these algorithms frequently outperform simpler models, explaining how individual predictions are produced can be challenging.

To improve transparency, SHAP (SHapley Additive Explanations) was implemented to quantify the contribution of each feature toward the final predicted vehicle price. Using SHAP, the model is able to identify which vehicle characteristics have the greatest influence on each prediction. This enables users to understand whether factors such as vehicle age, mileage, brand, fuel type, or ownership history increase or decrease the estimated asking price.

By providing interpretable explanations alongside predictions, SHAP transforms DriveValue AI from a purely predictive system into a transparent decision-support tool suitable for business environments where accountability and trust are essential.

Key Feature Importance

The SHAP analysis revealed that several vehicle characteristics consistently contributed more strongly to prediction performance than others. The most influential variables included:

- Vehicle Age
- Manufacturing Year
- Kilometres Driven
- Vehicle Brand
- Vehicle Model

These variables align with established principles of vehicle valuation, confirming that the model successfully learned realistic pricing relationships from the historical vehicle listings. Overall, the SHAP analysis demonstrated that the Random Forest model based its predictions on logical and commercially relevant vehicle characteristics, increasing confidence in the model's decision-making process.

Streamlit Deployment

To demonstrate real-world applicability, the trained Random Forest model was deployed as an interactive Streamlit web application. The application allows users to enter vehicle characteristics through an intuitive graphical interface, including:

- Manufacturing Year
- Vehicle Age
- Kilometres Driven
- Vehicle Brand
- Vehicle Model
- Fuel Type
- Transmission
- Ownership History
- Advertisement Posting Date

Once the required information is entered, the application processes the input using the same preprocessing pipeline applied during model training before generating an estimated asking price. This deployment demonstrates how the trained machine learning model can be used by dealerships, online vehicle marketplaces, or private sellers to obtain instant pricing recommendations without requiring technical knowledge of machine learning.

Production Workflow

The complete deployment architecture consists of sequential stages that transform raw vehicle information into an explainable price prediction. The workflow begins with vehicle information entered by the user through the Streamlit interface. The data is validated and transformed using the same preprocessing techniques applied during model development before being passed to the trained Random Forest model.

The model then generates an estimated asking price, after which SHAP provides an explanation highlighting the features that contributed most strongly to the prediction. Finally, the predicted vehicle

price and its explanation are presented to the end user through an intuitive interface. This workflow demonstrates how machine learning models can be deployed within dealership environments while maintaining both prediction accuracy and transparency.

Deployment Readiness

The completion of the DriveValue AI project demonstrates that the solution is technically prepared for deployment within a business environment. The project successfully achieved the following milestones:

- Development of an end-to-end machine learning pipeline.
- Training and evaluation of multiple regression algorithms.
- Selection of the best-performing production model.
- Integration of Explainable AI using SHAP.
- Model serialisation using Joblib for reuse.
- Development of a functional Streamlit web application.
- Preparation of a deployment workflow suitable for business implementation.

Collectively, these components demonstrate that DriveValue AI extends beyond academic experimentation by providing a practical and deployable intelligent vehicle pricing solution.

8. Conclusion

DriveValue AI successfully achieved its primary objective of developing an explainable machine learning framework capable of predicting used vehicle asking prices using historical vehicle listing data. By following the CRISP-DM methodology, the project progressed systematically through business understanding, data preparation, exploratory data analysis, feature engineering, predictive modelling, evaluation, explainability, and deployment planning.

A comprehensive data preparation process transformed the original dataset of 24,575 vehicle listings into a clean analytical dataset containing 14,988 high-quality observations represented by 466 engineered features. This preprocessing stage was fundamental in ensuring that the predictive models were trained using reliable and consistent information.

Three supervised regression algorithms were evaluated during model development, namely Linear Regression, Decision Tree Regressor, and Random Forest Regressor. Comparative evaluation demonstrated that the Random Forest Regressor consistently delivered the strongest predictive performance and was therefore selected as the final production model.

The final model achieved a Mean Absolute Error (MAE) of ₹398,156, a Root Mean Squared Error (RMSE) of ₹1,267,201, and an R^2 score of 0.446, indicating that the model successfully captured a substantial proportion of the variation in used vehicle asking prices while maintaining acceptable prediction accuracy on previously unseen data.

A distinguishing feature of DriveValue AI is the integration of Explainable Artificial Intelligence (SHAP). Rather than functioning solely as a predictive model, the solution provides interpretable explanations that identify the contribution of individual vehicle characteristics toward each predicted price. This significantly improves transparency, accountability, and stakeholder confidence when using machine learning to support pricing decisions.

The project was further strengthened through the development of a functional Streamlit web application, demonstrating that the trained model can be deployed within a real-world environment to generate instant pricing recommendations through an intuitive graphical user interface.

Overall, DriveValue AI demonstrates how machine learning can improve pricing consistency, reduce subjectivity, and support evidence-based decision-making within the automotive industry. The project provides a practical example of how predictive analytics and Explainable AI can be combined to deliver intelligent, transparent, and deployable business solutions.

9. Project Achievements

The following key objectives were successfully accomplished during the development of DriveValue AI:

- Developed a complete end-to-end machine learning solution following the CRISP-DM methodology.
- Cleaned and prepared a real-world dataset containing 24,575 vehicle listings.
- Engineered 466 predictive features from the original dataset.
- Evaluated multiple supervised regression algorithms.
- Selected the Random Forest Regressor as the final production model based on comparative performance.
- Integrated SHAP Explainable AI to improve model transparency and interpretability.
- Deployed the trained model using Streamlit to demonstrate real-world implementation.
- Created a structured GitHub repository documenting the complete project lifecycle.

These achievements demonstrate both the technical and practical success of the project while showcasing the complete machine learning lifecycle from problem identification to deployment.

10. Limitations

Although the project produced encouraging results, several limitations should be acknowledged.

- The dataset was sourced from a single online marketplace, meaning that pricing patterns may not fully represent broader automotive markets.
- Several potentially influential vehicle characteristics—such as accident history, service history, optional features, engine condition, and geographic location—were not available within the dataset. Including these variables could further improve prediction accuracy.

- While the Random Forest model performed well for the majority of vehicles, prediction errors were generally larger for luxury and premium vehicles because relatively few examples of these vehicles were available during training.
- Vehicle prices fluctuate continuously due to changing market conditions. Since the model was trained on historical listings, periodic retraining would be required to maintain predictive performance over time.

11. Recommendations

Several opportunities exist to further improve DriveValue AI and increase its value within commercial environments. Future development should focus on:

- Deploying the application to Streamlit Community Cloud, Microsoft Azure, or AWS to provide public access.
- Integrating live vehicle marketplace data to improve pricing relevance.
- Automating periodic model retraining to accommodate changing market conditions.
- Evaluating advanced ensemble algorithms such as XGBoost, LightGBM, and CatBoost to improve predictive accuracy.
- Incorporating additional vehicle attributes, including accident history, service records, engine specifications, and optional equipment.
- Developing an interactive dealership analytics dashboard for business reporting and pricing insights.

Implementing these enhancements would further improve scalability, prediction accuracy, and business value while supporting intelligent vehicle valuation within larger automotive organisations.